Combinatorics

external transform space

$$V_{24} = \left(\frac{4\pi}{5!}\right)\left(\frac{4\pi}{!5}\right)\left(\frac{4\pi}{35}\right)\left(\frac{4\pi}{18}\right)^2\left(\frac{4\pi}{32}\right)^3\left(\frac{4\pi}{8}\right)^4$$

$$\rho_{\text{Leech}} = \left(\frac{4\pi}{5!}\right)\left(\frac{4\pi}{!5}\right)\left(\frac{4\pi}{35}\right)\left(\frac{4\pi}{18}\right)^2\left(\frac{4\pi}{32}\right)^3\left(\frac{4\pi}{8}\right)^4$$

$5\left(\frac{4\pi}{2}\right)\left(\cos\left(\frac{7}{5}\right)\right)^2 = \frac{8}{e^{\phi_4}}$ mass

$2\left(\frac{5}{\sqrt{7}}\right)^2\left(\cos\left(\frac{5}{2}i\right)\right)^2\left(\cos\left(\frac{7}{5}\right)\right)^2 = \frac{32}{e^{\phi_3}}$ temperature

$\left(\frac{5}{\sqrt{7}}\right)\left(\frac{3^{-1/3}}{W_{We}}\right) = \frac{18}{e^{\phi_2}}$ charge

$\left(\sinh\left(\sinh\left(\frac{1}{7}\right)\right)\right)^{-1} = \frac{35}{e^{\phi_1}}$ length

$\left(\frac{4\pi}{4}\right)\left(\sinh\left(\frac{1}{4}\right)\right)^2 = \frac{!5}{e^{\phi_0}}$ time

$\phi_4 = 2.1764268381757\ldots$ $\quad 2(3-1)^2 = 8$
$\phi_3 = 1.4167869859079\ldots$ $\quad 2(3+1)^2 = 32$
$\phi_2 = 1.8755459671396\ldots$ $\quad 2(3\pm 0)^2 = 18$
$\phi_1 = 1.6162591817564\ldots$ $\quad 7\times 5 = 35 \quad \Gamma(5) = 24$
$\phi_0 = 5.3912583683231\ldots$ $\quad !5 = 44 \quad 5! = 120$

$m_p = 2.1764268381757\ldots\times 10^{-8}$ kg $\quad m_p = \phi_4\times 10^{-8}$ kg
$T_p = 1.4167869859079\ldots\times 10^{32}$ K $\quad T_p = \phi_3\times 10^{32}$ K
$q_p = 1.8755459671396\ldots\times 10^{-18}$ C $\quad q_p = \phi_2\times 10^{-18}$ C
$l_p = 1.6162591817564\ldots\times 10^{-35}$ m $\quad l_p = \phi_1\times 10^{-35}$ m
$t_p = 5.3912583683231\ldots\times 10^{-44}$ s $\quad t_p = \phi_0\times 10^{-44}$ s

$A_{ext} B_{ext}\,(1 + A_{int} \boxtimes) \qquad \boxtimes = \frac{l_p m_p}{q_p{}^2}\times\frac{\text{coulomb}^2}{\text{meter kilogram}}$

$$\frac{1}{x} + x + \frac{x^3}{2\pi} = (i^i)^{-\frac{4\pi}{8}} - \frac{m_p}{\text{kilogram}}$$

$\text{ж}_1 = 0.0854245431533304\ldots$
$\text{ж}_2 = 3.66756753485501\ldots$
$\text{ж}_3 = -1.87649603900417\ldots + 4.06615262615972\ldots\, i$
$\text{ж}_4 = -1.87649603900417\ldots - 4.06615262615972\ldots\, i$

$\text{ж}_r = 4.47826244916751\ldots$ $\quad \text{ж}_3 = \text{ж}_r e^{\text{ж}_\theta i}$
$\text{ж}_\theta = 2.00316562310924\ldots$ $\quad \text{ж}_4 = \text{ж}_r e^{-\text{ж}_\theta i}$

$\text{ж}_1\text{ж}_2\text{ж}_3\text{ж}_4 = 2\pi$ $\quad -Im(\text{ж}_3) = Im(\text{ж}_4)$
$\text{ж}_1+\text{ж}_2+\text{ж}_3+\text{ж}_4 = 0$ $\quad Im(\text{ж}_1) = 0$
$-\text{ж}_1-\text{ж}_2-\text{ж}_3-\text{ж}_4 = 0$ $\quad Im(\text{ж}_2) = 0$
$\text{ж}_1^2+\text{ж}_2^2+\text{ж}_3^2+\text{ж}_4^2 = -4\pi$
$-\text{ж}_1^2-\text{ж}_2^2-\text{ж}_3^2-\text{ж}_4^2 = 4\pi$ $\quad \frac{\text{ж}_1+\text{ж}_2}{\text{ж}_3+\text{ж}_4} = -1$
$\text{ж}_1+\text{ж}_2 = -2Re(\text{ж}_3) = -2Re(\text{ж}_4)$
$\text{ж}_1-\text{ж}_2 = Re(\text{ж}_1) - Re(\text{ж}_2)$ $\quad \frac{\text{ж}_1{}^2\text{ж}_2{}^2}{2\pi} = \frac{2\pi}{\text{ж}_3{}^2\text{ж}_4{}^2}$
$\text{ж}_3+\text{ж}_4 = 2Re(\text{ж}_3) = 2Re(\text{ж}_4)$
$\text{ж}_3-\text{ж}_4 = 2i\,Im(\text{ж}_3) = -2i\,Im(\text{ж}_4)$
$Re(\text{ж}_1)+Re(\text{ж}_2)+Re(\text{ж}_3)+Re(\text{ж}_4) = 0$
$Im(\text{ж}_1)+Im(\text{ж}_2)+Im(\text{ж}_3)+Im(\text{ж}_4) = 0$
$Re(\text{ж}_3)^2+Re(\text{ж}_4)^2 = \frac{1}{2}(\text{ж}_1+\text{ж}_2)^2$
$Im(\text{ж}_3)^2+Im(\text{ж}_4)^2 = \frac{1}{2}(\text{ж}_3-\text{ж}_4)^2$
$\text{ж}_1\text{ж}_2+\text{ж}_3\text{ж}_4+((\text{ж}_1+\text{ж}_2)(\text{ж}_3+\text{ж}_4)) = 2\pi$
$\text{ж}_1\text{ж}_2+\text{ж}_1\text{ж}_3+\text{ж}_1\text{ж}_4+\text{ж}_2\text{ж}_3+\text{ж}_2\text{ж}_4+\text{ж}_3\text{ж}_4 = 2\pi$
$Re(\text{ж}_3)Re(\text{ж}_4) - Im(\text{ж}_3)Im(\text{ж}_4) = \text{ж}_r{}^2$
$Re(\text{ж}_3)^2+Im(\text{ж}_3)^2 = \text{ж}_r{}^2$ $\quad \text{ж}_3\text{ж}_4 = \text{ж}_r{}^2$
$Re(\text{ж}_4)^2+Im(\text{ж}_4)^2 = \text{ж}_r{}^2$ $\quad \text{ж}_1\text{ж}_2 = \frac{2\pi}{\text{ж}_r{}^2}$
$Re(\text{ж}_3)^2+Im(\text{ж}_4)^2 = \text{ж}_r{}^2$
$Re(\text{ж}_4)^2+Im(\text{ж}_3)^2 = \text{ж}_r{}^2$
$(a^2+b^2)(c^2+d^2) = (ac-bd)^2+(ad+bc)^2 = \text{ж}_r{}^4$
$(a^2+b^2)(c^2+d^2) = (ac+bd)^2+(ad-bc)^2 = \text{ж}_r{}^4$
$a = Re(\text{ж}_3)\quad b = Im(\text{ж}_3)\quad c = Re(\text{ж}_4)\quad d = Im(\text{ж}_4)$
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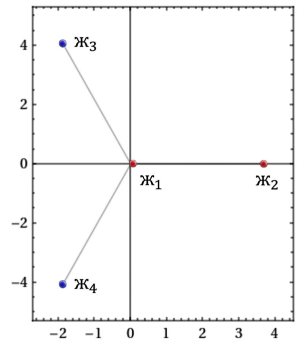
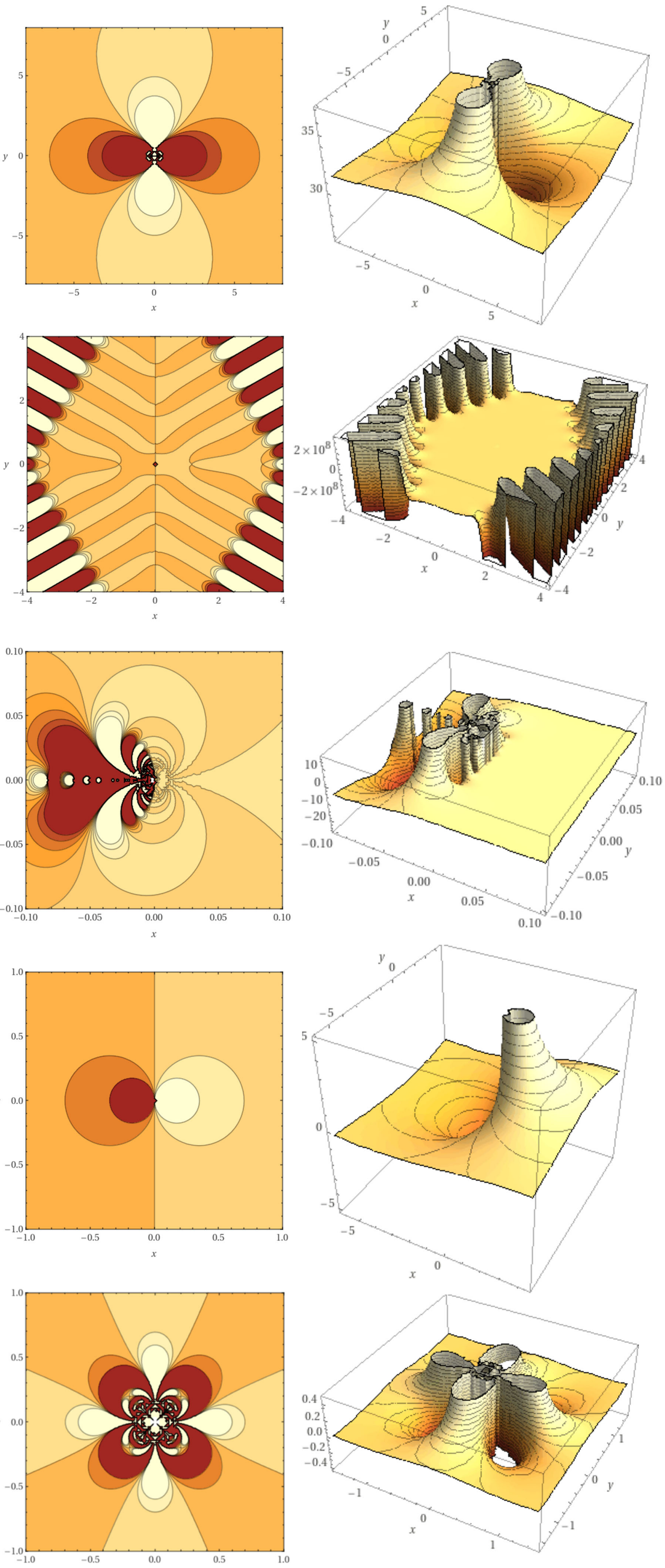


$\sum_{n=1}^{4}\text{ж}_n = 0 \qquad \prod_{n=1}^{4}\text{ж}_n = 2\pi \qquad \sum_{n=1}^{4}\text{ж}_n{}^2 = -4\pi$



$\log((i^i)^0) = \text{ж}_1+\text{ж}_2+\text{ж}_3+\text{ж}_4$
$\log((i^i)^4) = -\text{ж}_1\text{ж}_2\text{ж}_3\text{ж}_4$
$\log((i^i)^8) = \text{ж}_1{}^2+\text{ж}_2{}^2+\text{ж}_3{}^2+\text{ж}_4{}^2$

$S_{n-1}(r) = \frac{2\pi^{n/2}}{\Gamma\left(\frac{n}{2}\right)}r^{n-1}$ $\qquad V_n(r) = \frac{\pi^{n/2}}{\Gamma\left(\frac{n}{2}+1\right)}r^n$

$\log((i^i)^4) = -1\,\text{ж}_1\text{ж}_2\text{ж}_3\text{ж}_4$
$\log((i^i)^8) = -2\,\text{ж}_1\text{ж}_2\text{ж}_3\text{ж}_4$
$\log((i^i)^{12}) = -3\,\text{ж}_1\text{ж}_2\text{ж}_3\text{ж}_4$
$\log((i^i)^{16}) = -4\,\text{ж}_1\text{ж}_2\text{ж}_3\text{ж}_4$

$\frac{S_{n+1}(r)}{V_n(r)} = 2\pi r \qquad \frac{S_{n-1}(r)}{V_n(r)} = \frac{n}{r}$

$\log((i^i)^8) = 1\,(\text{ж}_1{}^2+\text{ж}_2{}^2+\text{ж}_3{}^2+\text{ж}_4{}^2)$
$\log((i^i)^{16}) = 2\,(\text{ж}_1{}^2+\text{ж}_2{}^2+\text{ж}_3{}^2+\text{ж}_4{}^2)$
$\log((i^i)^{24}) = 3\,(\text{ж}_1{}^2+\text{ж}_2{}^2+\text{ж}_3{}^2+\text{ж}_4{}^2)$
$\log((i^i)^{32}) = 4\,(\text{ж}_1{}^2+\text{ж}_2{}^2+\text{ж}_3{}^2+\text{ж}_4{}^2)$

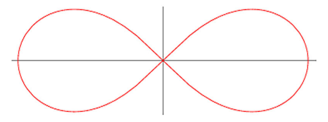
$(i^i)^{-1} = \frac{s^4 W_{We}{}^4}{8}$, $(i^i)^{-2} = e^\pi$, $(i^i)^{-4} = e^{2\pi}$, $(i^i)^{-6} = e^{3\pi}$, $(i^i)^{-8} = e^{4\pi}$

$$\sum_{n=0}^{\infty}\frac{1}{n!} = \frac{1}{0!}+\frac{1}{1!}+\frac{1}{2!}+\frac{1}{3!}+\frac{1}{4!}+\frac{1}{5!}+\frac{1}{6!}+\frac{1}{7!}+\frac{1}{8!}+\cdots = e$$
$$\sum_{n=0}^{\infty}\frac{(-1)^n}{n!} = \frac{1}{0!}-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}-\frac{1}{5!}+\frac{1}{6!}-\frac{1}{7!}+\frac{1}{8!}-\cdots = \frac{1}{e}$$
$$\sum_{n=0}^{\infty}x^n = x^0+x^1+x^2+x^3+x^4+x^5+x^6+x^7+\cdots = \frac{1}{1-x}$$
$$\sum_{n=0}^{\infty}(-x)^n = x^0-x^1+x^2-x^3+x^4-x^5+x^6-x^7+\cdots = \frac{1}{1+x}$$

$\left[Li_2\left(\frac{1}{\varphi_i}\right)+Li_2(\varphi_i)\right] = \frac{(4\pi)^2}{288}$
$i\left[Li_2\left(\frac{1}{\varphi_i}\right)-Li_2(\varphi_i)\right] = V_{fe}$
$i\left[Li_2\left(\frac{1}{\varphi_i{}^2}\right)-Li_2(\varphi_i{}^2)\right] = \frac{2V_{fe}}{3}$
$i\left[Li_2\left(\frac{1}{\varphi_i{}^3}\right)-Li_2(\varphi_i{}^3)\right] = 0$
$i\left[Li_2\left(\frac{1}{\varphi_i{}^4}\right)-Li_2(\varphi_i{}^4)\right] = -\frac{2V_{fe}}{3}$
$i\left[Li_2\left(\frac{1}{\varphi_i{}^5}\right)-Li_2(\varphi_i{}^5)\right] = -V_{fe}$
$i\left[Li_2\left(\frac{1}{\varphi_i{}^6}\right)-Li_2(\varphi_i{}^6)\right] = 0$
$i\left[Li_2\left(\frac{1}{i}\right)-Li_2(i)\right] = 2K$
$i\left[Li_2\left(\frac{1}{i^2}\right)-Li_2(i^2)\right] = 0$
$i\left[Li_2\left(\frac{1}{i^3}\right)-Li_2(i^3)\right] = -2K$
$i\left[Li_2\left(\frac{1}{i^4}\right)-Li_2(i^4)\right] = 0$
$\left[Li_2\left(\frac{1}{i}\right)+Li_2(i)\right] = -\frac{1}{6}\left(\frac{4\pi}{8}\right)^2$
$Li_2(1) - Li_2(-1) = \left(\frac{4\pi}{8}\right)^2$
$Li_2(-1) = -\left(\frac{4\pi}{\Gamma(5)}\right)\left(\frac{4\pi}{8}\right)$
$Li_2(1) = \left(\frac{4\pi}{\Gamma(5)}\right)\left(\frac{4\pi}{4}\right)$
$\sum_{n=1}^{\infty}\frac{1}{n^2} = Li_2(1)$
$\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2} = Li_2(-1)$
$\sum_{n=1}^{\infty}\frac{1}{(2n+1)^2} = \left(\frac{4\pi}{16}\right)\left(\frac{4\pi}{8}\right)$
$\sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)^2} = K$
$\sum_{n=1}^{\infty}\frac{\sin\left(n\frac{\pi}{2}\right)}{n^2} = K$
$\sum_{n=1}^{\infty}\frac{\sin\left(n\frac{\pi}{3}\right)}{n^2} = G_{Gi}$
$\sum_{n=1}^{\infty}\frac{\cos\left(n\frac{\pi}{3}\right)}{n^2} = \left(\frac{4\pi}{\Gamma(5)}\right)^2$
$\sum_{n=1}^{\infty}\frac{\cos\left(n\frac{\pi}{2}\right)}{n^2} = -\left(\frac{4\pi}{\Gamma(5)}\right)\left(\frac{4\pi}{32}\right)$
$\sum_{n=0}^{\infty}\binom{n}{k} = 2^n$
$\sum_{n=1}^{\infty}\frac{1}{n^s} = \zeta(s) \qquad \sum_{n=0}^{\infty}\frac{H_n}{n^2} = 2\zeta(3)$
$\sum_{n=0}^{\infty}\varphi^{-n} = \varphi^2 \qquad \sum_{n=0}^{\infty}\psi^{-n} = \psi^3$
$\sum_{n=1}^{\infty}\frac{F_n}{F_{2n}} = \mathcal{L}_{RL}$
$\sum_{n=1}^{\infty}\frac{1}{F_n} = \mathcal{F} = \sqrt{5}\sum_{n=1}^{\infty}\frac{1}{\varphi^n-(-\varphi)^n}$
$\sum_{n=1}^{\infty}\frac{1}{L_n} = \mathcal{L}_{RL} = \sum_{n=1}^{\infty}\frac{1}{(-\varphi)^{-n}+\varphi^n}$
$\sum_{n=0}^{\infty}\frac{x^n}{n!} = \frac{x^0}{0!}+\frac{x^1}{1!}+\frac{x^2}{2!}+\cdots = e^x$

lemniscate

$(x^2+y^2)^2 = (x^2-y^2)$



$|z^2+1| = |z^2-1|$

defines points equidistant from roots 0, 1, and −1

$Li_2(\varphi_i) = \left(\frac{4\pi}{\Gamma(5)}\right)^2 + G_{Gi}\,i$
$Li_2\left(\frac{1}{\varphi_i}\right) = \left(\frac{4\pi}{\Gamma(5)}\right)^2 - G_{Gi}\,i$
$Li_2(i) = -\left(\left(\frac{4\pi}{\Gamma(5)}\right)\left(\frac{4\pi}{32}\right)\right) + K\,i$
$Li_2\left(\frac{1}{i}\right) = -\left(\left(\frac{4\pi}{\Gamma(5)}\right)\left(\frac{4\pi}{32}\right)\right) - K\,i$
$\log\left((i^i)^{-\pi/2}\right) = \left(\frac{4\pi}{8}\right)^2$
$\log\left((i^i)^{-32\pi}\right) = (4\pi)^2$
$\log\left((i^i)^8\right) = -4\pi$
$\log\left((i^i)^4\right) = -2\pi$
$\log\left((i^i)^2\right) = -\left(\frac{4\pi}{4}\right)$
$\log\left((i^i)^{1/1}\right) = -\left(\frac{4\pi}{8}\right)$
$\log\left((i^i)^{1/2}\right) = -\left(\frac{4\pi}{16}\right)$
$\log\left((i^i)^{1/3}\right) = -\left(\frac{4\pi}{\Gamma(5)}\right)$
$\log\left((i^i)^{1/4}\right) = -\left(\frac{4\pi}{32}\right)$
$\frac{\mathcal{S}}{2} = \sin\left(\frac{3\pi}{14}\right)+1$
$\frac{\mathcal{S}}{2} = \cos\left(\frac{4\pi}{14}\right)+1$
$e^{\pi i} = -1$
$e^\pi = (i^{-i})^2$
$i^{-i} = \frac{s^4 W_{We}{}^4}{8}$
$e^\pi = \frac{s^8 W_{We}{}^8}{8^2}$
$i^{-i} = 2L^4 W_{We}{}^4$
$e^\pi = 4L^8 W_{We}{}^8$
$(i^i)^{-4\pi/8} = \frac{s^{2\pi}W_{We}{}^{2\pi}}{\sqrt{8^\pi}}$
$(i^i)^{-2\pi} = \frac{s^{8\pi}W_{We}{}^{8\pi}}{8^{2\pi}}$
$W_{We} = \frac{\sqrt{\sqrt{8}}}{s}e^{\left(\frac{4\pi}{32}\right)}$
$W_{We} = \frac{1}{L}\left(\frac{i^{-i}}{2}\right)^{1/4}$
$W_{We} = \frac{1}{L}\left(\frac{e^\pi}{4}\right)^{1/8}$
$W_{We} = \frac{1}{L}\left(\frac{e^{2\pi}}{16}\right)^{1/16}$
$W_{We} = \frac{1}{L}\left(\frac{e^{4\pi}}{8\cdot 32}\right)^{1/32}$
$e^{\zeta'(0)} = \frac{1}{\sqrt{2\pi}}$
$e^{\beta'(0)} = \frac{s}{\sqrt{4\pi}}$
$\omega_1 = \frac{\Gamma^3\left(\frac{1}{3}\right)}{4\pi}\varphi_i$
$\omega_2 = \frac{\Gamma^3\left(\frac{1}{3}\right)}{4\pi}$
$s = \frac{\Gamma^2\left(\frac{1}{4}\right)}{\sqrt{2\pi}}$
$L = \frac{\Gamma^2\left(\frac{1}{4}\right)}{\sqrt{8\pi}}$
$L_1 = \frac{\Gamma^2\left(\frac{1}{4}\right)}{\sqrt{32\pi}}$
$L_2 = \frac{2\pi}{\sqrt{2}}\frac{\Gamma\left(\frac{1}{2}\right)}{\Gamma^2\left(\frac{1}{4}\right)}$
$C_U = 2\pi\frac{\Gamma\left(\frac{1}{2}\right)}{\Gamma^2\left(\frac{1}{4}\right)}$
$G_{Ga} = \frac{\Gamma^2\left(\frac{1}{4}\right)}{2\pi\sqrt{2\pi}}$

$\frac{r_n}{r_e} = \left(\frac{4\pi}{\Gamma(5)}\right)^2$
$\frac{r_n}{r_+} = K$
$G_{Gi} = V_{\text{ideal tetrahedron}}$
$2G_{Gi} = V_{fe}$
$2K = 2\pi C_d$
$4K = 2\pi\log(D_d)$
$4K = V_{\text{Whitehead link}}$
$4K = V_{\text{ideal octahedron}}$
$8K = V_{\text{Borromean rings}}$

$\sin\left(\frac{3\pi}{14}\right) = \cos\left(\frac{4\pi}{14}\right)$
$\sin\left(\frac{4\pi}{14}\right) = \cos\left(\frac{3\pi}{14}\right)$
$\left(\frac{4\pi}{8}\right) = L\,L_2$
$\left(\frac{4\pi}{16}\right) = L_1 L_2 \qquad \cos\left(\frac{4\pi}{8}\right) = 0$
$\left(\frac{4\pi}{32}\right) = \frac{L\,L_1 L_2}{s} \qquad \sin\left(\frac{4\pi}{8}\right) = 1$

$\int_0^{2\pi/12} -4\log|e^{-xi}-e^{xi}| = V_{fe}$
$\int_0^{2\pi/6} -6\log|e^{-xi}-e^{xi}| = V_{fe}$
$\int_0^{1/2}\Gamma(1+x)\,\Gamma(1-x)\,dx = \frac{4K}{2\pi}$
$\int_0^1\Gamma(1+x)\,\Gamma(1-x)\,dx = \frac{8K}{2\pi}$
$\int_0^1\int_0^1\frac{1}{1+x^2y^2} = K$
$\int_{-\infty}^{\infty}e^{-x^2}dx = \sqrt{\frac{1}{2}2\pi}$
$\int_{-\infty}^{\infty}e^{-x^4}dx = \sqrt{\frac{L}{2}\sqrt{2\pi}}$
$\int_{-1}^{1}\sqrt{1+x^2}\,dx = P_{up}$
$\int_{-1}^{1}\sqrt{1-x^2}\,dx = \left(\frac{4\pi}{8}\right)$
$\int_{-\infty}^{\infty}\frac{1}{1+x^2}dx = \pi$
$\int_{-\infty}^{\infty}\frac{1}{1+x^4}dx = \frac{4\pi}{\sqrt{32}}$
$\int_0^1\frac{1}{\sqrt{1+x^4}}dx = \frac{s}{\sqrt{32}}$
$\int_0^1\frac{1}{\sqrt{1-x^4}}dx = \frac{s}{4}$
$\int_0^1\sqrt{1-x^4}\,dx = \frac{s}{6}$
$\int_0^1\frac{x^2}{\sqrt{1-x^4}}dx = \frac{\pi}{s}$
$\int_0^{1/2}\int_0^x 6\sqrt{x^2+y^2}\,dy\,dx = \frac{P_{up}}{8}$
$\int_0^\infty 2^{-x}dx = \frac{1}{\log(2)}$
$\int_0^1\frac{\log(x)}{e^x}dx = \gamma$
$\int_0^\infty\log\left(\log\left(\frac{1}{x}\right)\right)dx = \gamma$

$V_{\text{unit sphere}} = \left(\frac{4\pi}{3}\right)$
$V_{\text{Gabriel's horn}} = \left(\frac{4\pi}{4}\right)$
$V_{\text{pesudosphere}} = \left(\frac{4\pi}{6}\right)$
$V_{\text{exp trumpet}} = \left(\frac{4\pi}{8}\right)$
$V_{\text{cos spindle}} = \left(\frac{4\pi}{4}\right)\left(\frac{4\pi}{8}\right)$

$A_{\text{one pedal}} = 1$
$A_{\text{unit lemniscate}} = 2$
$A_{\text{exp trumpet}} = \pi P_{up}$
$A_{\text{cos spindle}} = 2\pi P_{up}$
$A_{\text{unit circle}} = 2\pi$
$A_{\text{unit sphere}} = 4\pi$
$A_{\text{pesudosphere}} = 4\pi$
$A_{\text{unit cardioid}} = 6\pi$

$L_{\text{unit wave}} = \frac{4\pi}{s}+s$
$L_{\text{unit cardioid}} = 8$
$L_{\text{unit cardioid cusp normal}} = \sqrt{\frac{32}{18}}$

$\alpha_F = \lim_{n\to\infty}\frac{d_n}{d_{n+1}}$
$\delta_F = \lim_{n\to\infty}\frac{a_{n-1}-a_{n-2}}{a_n-a_{n-1}}$

3D sphere packing densities

$\rho_{\text{cubic lat}} = \left(\frac{4\pi}{\Gamma(5)}\right)$
$\rho_{\text{hex lat}} = \frac{2}{\sqrt{3}}\left(\frac{4\pi}{\Gamma(5)}\right)$
$\rho_{\text{tet lat}} = \frac{\sqrt{3}}{2}\left(\frac{4\pi}{32}\right)$
$\rho_{BCC} = \sqrt{3}\left(\frac{4\pi}{32}\right)$
$\rho_{\min} = \frac{1}{\sqrt{2}}\left(\frac{4\pi}{18}\right)$
$\rho_{FCC} = \frac{4\pi}{\sqrt{288}}$
$\rho_{HCP} = \frac{4\pi}{\sqrt{288}}$

densest hypersphere lattice packings

$\delta_2 = \sqrt{3}\left(\frac{4\pi}{\Gamma(5)}\right)$
$\delta_3 = \sqrt{2}\left(\frac{4\pi}{\Gamma(5)}\right)$
$\delta_4 = \left(\frac{4\pi}{16}\right)^2$
$\delta_5 = \sqrt{2}\left(\frac{4\pi}{5!}\right)\left(\frac{4\pi}{4}\right)$
$\delta_6 = \sqrt{3}\left(\frac{4\pi}{16}\right)\left(\frac{4\pi}{18}\right)\left(\frac{4\pi}{32}\right)$
$\delta_7 = \left(\frac{4\pi}{\Gamma(5)}\right)\left(\frac{4\pi}{35}\right)\left(\frac{4\pi}{8}\right)$
$\delta_8 = \left(\frac{4\pi}{\Gamma(5)}\right)\left(\frac{4\pi}{16}\right)^3$

max density sphere packings

$\rho_{2D} = \sqrt{\left(\frac{4\pi}{\Gamma(5)}\right)\left(\frac{4\pi}{8}\right)}$
$\rho_{3D} = \frac{4\pi}{\sqrt{288}}$
$\rho_{4D} = \left(\frac{4\pi}{16}\right)^2$
$\rho_{8D} = \left(\frac{4\pi}{\Gamma(5)}\right)\left(\frac{4\pi}{16}\right)^3$

factors of 288

1	288
2	144
3	96
4	72
6	48
8	**36**
9	**32**
12	24
16	**18**

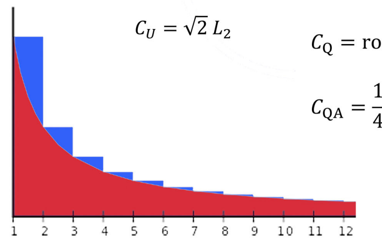
$\Gamma(\varphi+1) = \Gamma(\varphi-1)$
$\Gamma(\varphi_i+1) = -\Gamma(\varphi_i-1)$
$\Gamma\left(-\frac{1}{\varphi}+1\right) = \Gamma\left(-\frac{1}{\varphi}-1\right)$
$\Gamma\left(\frac{1}{\varphi_i}+1\right) = -\Gamma\left(\frac{1}{\varphi_i}-1\right)$

$\zeta\left(\tan\left(\frac{4\pi}{32}\right)+1\right) = \zeta\left(\cot\left(\frac{4\pi}{32}\right)-1\right)$
$\zeta\left(-\tan\left(\frac{4\pi}{32}\right)-1\right) = \zeta\left(-\cot\left(\frac{4\pi}{32}\right)+1\right)$

$\prod_{n=1}^{\infty}\left(1-\left(-\frac{1}{\varphi^2}\right)^n\right) = F_{FF}$
$\prod_{p\text{ prime}}\left(\frac{p^2+1}{p^2-1}\right) = \frac{5}{2}$
$\prod_{p\text{ prime}}\left(\frac{p^4+1}{p^4-1}\right) = \frac{7}{6}$
$\prod_{p\text{ prime}}\left(\frac{p^s+1}{p^s-1}\right) = \frac{\zeta(s)^2}{\zeta(2s)}$
$\prod_{p\text{ prime}}\left(1+\frac{1}{p^s}\right) = \frac{\zeta(s)}{\zeta(2s)}$
$\prod_{p\text{ prime}}\left(1-\frac{1}{p^s}\right) = \frac{1}{\zeta(s)}$
$\prod_{p\text{ prime}}\left(1+\frac{1}{p(p-1)}\right) = \left(\frac{35}{4\pi}\right)\left(\frac{18}{4\pi}\right)\left(\frac{8}{4\pi}\right)^2\zeta(3)$

$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$
$\Gamma\left(-\frac{1}{2}\right) = -\sqrt{4\pi}$
$\Gamma\left(\frac{1}{2}\right)\Gamma\left(-\frac{1}{2}\right) = -2\pi$
$\Gamma\left(\frac{3}{2}\right) = \sqrt{\frac{4\pi}{16}}$
$B\left(\frac{1}{2},\frac{3}{2}\right) = \left(\frac{4\pi}{8}\right)$
$B\left(\frac{1}{2},\frac{1}{2}\right) = \pi$
$B\left(\frac{1}{2},\frac{1}{4}\right) = s$

$\lim_{n\to\infty}\frac{1}{\log(p_n)}\prod_{k=1}^{n}\frac{1}{1-\frac{1}{p_k}} = e^\gamma$
$\lim_{n\to\infty}\log(p_n)\prod_{k=1}^{n}\frac{1}{1+\frac{1}{p_k}} = \frac{6}{e^\gamma}\left(\frac{4\pi}{\Gamma(5)}\right)^2$
$\lim_{x\to 0}\frac{1}{x}\left(\frac{1}{\Psi(1-x)}-\frac{1}{\Psi(1+x)}\right) = \frac{6}{\gamma^2}\frac{(4\pi)^2}{288}$
$\lim_{n\to\infty}\frac{\pi^0}{\Gamma(1)}+\frac{\pi^1}{\Gamma(2)}+\frac{\pi^2}{\Gamma(3)}+\cdots+\frac{\pi^n}{\Gamma(n+1)} = e^\pi$
$\lim_{x\to 0}\frac{1}{x}\left(\frac{1}{\Gamma(1+x)}-\frac{1}{\Gamma(1-x)}\right) = 2\gamma$
$\lim_{x\to 0}\left(\frac{\zeta(1+x)+\zeta(1-x)}{2}\right) = \gamma$
$\lim_{n\to\infty}\left(\sum_{k=1}^{n}\frac{1}{k}-\int_1^n\frac{1}{x}dx\right) = \gamma$
$\Gamma'(1) = -\gamma$



$s = 2L$
$L = 2L_1$
$L_1 = \frac{s}{4}$
$L_2 = \frac{\pi}{s}$
$G_{Ga} = \frac{s}{2\pi}$
$C_U = \sqrt{2}L_2$

$s = 4K(i)$
$C_d = \frac{2K}{2\pi}$
$D_d = e^{\left(\frac{4K}{2\pi}\right)}$
$D_d = e^{2C_d}$
$S = \log\left((e^\gamma C_U)^2\right)$
$C_Q = \text{root of }\zeta\left(\frac{1}{2},\frac{x}{2}\right)$
$C_{QA} = \frac{1}{4}\zeta\left(\frac{3}{2},\frac{C_Q}{2}\right)$

fixed point equation for $i^{i^{i^{\cdots}}}$

$x = i^x$
$x = \cos\left(\frac{4\pi}{8}x\right)+i\sin\left(\frac{4\pi}{8}x\right)$
$\log(x) = \left(\frac{4\pi}{8}\right)xi$
$x = e^{\left(\frac{4\pi}{8}\right)xi}$

fixed point equation for $\pm C_{CFP}$

$e^x(x-1) = e^{-x}(x+1)$
$x = \frac{e^x+e^{-x}}{e^x-e^{-x}}$
$\coth(x) = x$
$\operatorname{csch}(C_{CFP}) = L_{LL}$
$\operatorname{csch}(-C_{CFP}) = -L_{LL}$
$\sinh(C_{CFP}) = \frac{1}{L_{LL}}$
$\sinh(-C_{CFP}) = -\frac{1}{L_{LL}}$

fixed point equation for D_{Do}

$\cos(x) = x$
$x = Re(e^{xi})$

fixed point equation for Ω

$x = e^{-x}$
$x = -\log(x)$
$x = \log\left(\frac{1}{x}\right)$

$(1+i)!(1-i)! = \frac{4\pi}{e^\pi-e^{-\pi}}$
$(0+i)!(0-i)! = \frac{2\pi}{e^\pi-e^{-\pi}}$
$|\Gamma(0+i)|^2 = \frac{2\pi}{e^\pi-e^{-\pi}}$
$\left|\Gamma\left(\frac{1}{2}+i\right)\right|^2 = \frac{2\pi}{e^\pi+e^{-\pi}}$
$|\Gamma(1+i)|^2 = \frac{2\pi}{e^\pi-e^{-\pi}}$

$\varphi = 1+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{1+\ddots}}}}}$

$\frac{1}{\varphi} = 0+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{1+\cfrac{1}{1+\ddots}}}}}$

$C_{CF} = 0+\cfrac{1}{1+\cfrac{1}{2+\cfrac{1}{3+\cfrac{1}{4+\cfrac{1}{5+\ddots}}}}}$

$C_{R1} = 0+\cfrac{e^{-0\pi}}{1+\cfrac{e^{-2\pi}}{1+\cfrac{e^{-4\pi}}{1+\cfrac{e^{-6\pi}}{1+\cfrac{e^{-8\pi}}{1+\ddots}}}}}$

$\frac{1}{1-e} = 0+\cfrac{1}{1+\cfrac{2}{2+\cfrac{3}{3+\cfrac{4}{4+\cfrac{5}{5+\ddots}}}}}$

$e = 2+\cfrac{1}{1+\cfrac{1}{2+\cfrac{2}{3+\cfrac{3}{4+\cfrac{4}{5+\ddots}}}}}$

$\left(\frac{4\pi}{8}\right) = 1+\cfrac{1}{\frac{1}{1}+\cfrac{1}{\frac{1}{2}+\cfrac{1}{\frac{1}{3}+\cfrac{1}{\frac{1}{4}+\cfrac{1}{\frac{1}{5}+\ddots}}}}}$

$\left(\frac{4\pi}{16}\right) = 0+\cfrac{1}{1+\cfrac{1}{3+\cfrac{1}{5+\cfrac{1}{7+\cfrac{1}{9+\ddots}}}}}$

$\left(\frac{16}{4\pi}\right) = 1+\cfrac{1^2}{2+\cfrac{3^2}{2+\cfrac{5^2}{2+\cfrac{7^2}{2+\cfrac{9^2}{2+\ddots}}}}}$

$\left(\frac{s^2}{4\pi}\right) = 2+\cfrac{1^2}{4+\cfrac{3^2}{4+\cfrac{5^2}{4+\cfrac{7^2}{4+\cfrac{9^2}{4+\ddots}}}}}$

$\left(\frac{4\pi}{4}\right) = 3+\cfrac{1^2}{6+\cfrac{3^2}{6+\cfrac{5^2}{6+\cfrac{7^2}{6+\cfrac{9^2}{6+\ddots}}}}}$

$18\left(\frac{2\pi}{s^2}\right) = 4+\cfrac{1^2}{8+\cfrac{3^2}{8+\cfrac{5^2}{8+\cfrac{7^2}{8+\cfrac{9^2}{8+\ddots}}}}}$

$K = 0+\cfrac{1^4}{1+\cfrac{1^4}{8+\cfrac{3^4}{16+\cfrac{5^4}{24+\cfrac{7^4}{32+\ddots}}}}}$

$2K = 1+\cfrac{1\cdot 1}{3+\cfrac{1\cdot 2}{1+\cfrac{2\cdot 2}{3+\cfrac{2\cdot 3}{1+\cfrac{3\cdot 3}{3+\ddots}}}}}$

$\frac{\pi}{2} = 1+\cfrac{1\cdot 1}{1+\cfrac{1\cdot 2}{1+\cfrac{2\cdot 3}{1+\cfrac{3\cdot 4}{1+\cfrac{4\cdot 5}{1+\ddots}}}}}$

$\frac{L}{2} = 1+\cfrac{1\cdot 1}{2+\cfrac{2\cdot 3}{2+\cfrac{4\cdot 5}{2+\cfrac{6\cdot 7}{2+\cfrac{8\cdot 9}{2+\ddots}}}}}$

$\cosh(x) = \frac{1}{2}(e^x+e^{-x})$
$\sinh(x) = \frac{1}{2}(e^x-e^{-x})$
$\cos(x) = \frac{1}{2}(e^{xi}+e^{-xi})$
$\sin(x) = \frac{i}{2}(e^{xi}-e^{-xi})$
$\cosh(x) = \frac{x^0}{0!}+\frac{x^2}{2!}+\frac{x^4}{4!}+\frac{x^6}{6!}+\frac{x^8}{8!}+\cdots$
$\sinh(x) = \frac{x^1}{1!}+\frac{x^3}{3!}+\frac{x^5}{5!}+\frac{x^7}{7!}+\frac{x^9}{9!}+\cdots$
$\cos(x) = \frac{x^0}{0!}-\frac{x^2}{2!}+\frac{x^4}{4!}-\frac{x^6}{6!}+\frac{x^8}{8!}-\cdots$
$\sin(x) = \frac{x^1}{1!}-\frac{x^3}{3!}+\frac{x^5}{5!}-\frac{x^7}{7!}+\frac{x^9}{9!}-\cdots$
$\tan^{-1}(x) = \frac{x^1}{1}-\frac{x^3}{3}+\frac{x^5}{5}-\frac{x^7}{7}+\frac{x^9}{9}-\cdots$
$\log(1+x) = \frac{x^1}{1}-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\frac{x^5}{5}-\cdots$
$\log(1-x) = -\frac{x^1}{1}-\frac{x^2}{2}-\frac{x^3}{3}-\frac{x^4}{4}-\frac{x^5}{5}-\cdots$
$\log(x) = \frac{x^{-1}}{1}-\frac{x^{-2}}{2}+\frac{x^{-3}}{3}-\frac{x^{-4}}{4}+\frac{x^{-5}}{5}-\cdots$
$Li_s(x) = \frac{x^1}{1^s}+\frac{x^2}{2^s}+\frac{x^3}{3^s}+\frac{x^4}{4^s}+\frac{x^5}{5^s}+\cdots$
$\zeta(1) = \frac{1}{1}+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\frac{1}{5}+\cdots = \infty$
$\zeta(2) = \frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+\frac{1}{4^2}+\frac{1}{5^2}+\cdots = \frac{\pi^2}{6}$
$\zeta(3) = \frac{1}{1^3}+\frac{1}{2^3}+\frac{1}{3^3}+\frac{1}{4^3}+\frac{1}{5^3}+\cdots$
$\zeta(3) = \frac{1}{2}\int_0^\infty\frac{\log(x)\log(1-x)}{x(1-x)}dx$
$\zeta(3) = \frac{1}{2}\int_0^\infty\frac{x^2}{e^x-1}dx \qquad \sqrt{\frac{\zeta(3)}{\zeta'(-2)}} = 2\pi i$
$\zeta(3) = \frac{2}{3}\int_0^\infty\frac{x^2}{e^x+1}dx \qquad \frac{\zeta\left(\frac{3}{2}\right)}{\zeta\left(-\frac{1}{2}\right)} = -4\pi$

Pascal's triangle

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1
1 6 15 20 15 6 1
1 7 21 35 35 21 7 1

$\sqrt{288} = 16+\cfrac{1}{1+\cfrac{1}{32+\cfrac{1}{1+\cfrac{1}{32+\cfrac{1}{1+\ddots}}}}}$

natural numbers $= \binom{n+0}{1} = \frac{n}{1!}$
triangular numbers $= \binom{n+1}{2} = \frac{n(n+1)}{2!}$
tetrahedral numbers $= \binom{n+2}{3} = \frac{n(n+1)(n+2)}{3!}$
pentatope numbers $= \binom{n+3}{4} = \frac{n(n+1)(n+2)(n+3)}{4!}$
5 − simplex numbers $= \binom{n+4}{5} = \frac{n(n+1)(n+2)(n+3)(n+4)}{5!}$

$1^2+2^2+3^2+4^2+5^2+\cdots+n^2 = \frac{n(n-1)(2n+1)}{3!}$
$1^3+2^3+3^3+4^3+5^3+\cdots+n^3 = (1+2+3+4+5+\cdots+n)^2$
$(1+x)^n = \frac{n^{\underline{0}}x^0}{0!}+\frac{n^{\underline{1}}x^1}{1!}+\frac{n^{\underline{2}}x^2}{2!}+\frac{n^{\underline{3}}x^3}{3!}+\frac{n^{\underline{4}}x^4}{4!}+\cdots+\frac{n^{\underline{n}}x^n}{n!}$
$2^n = \frac{n^{\underline{0}}}{0!}+\frac{n^{\underline{1}}}{1!}+\frac{n^{\underline{2}}}{2!}+\frac{n^{\underline{3}}}{3!}+\frac{n^{\underline{4}}}{4!}+\frac{n^{\underline{5}}}{5!}+\frac{n^{\underline{6}}}{6!}+\frac{n^{\underline{7}}}{7!}+\frac{n^{\underline{8}}}{8!}+\frac{n^{\underline{9}}}{9!}+\frac{n^{\underline{10}}}{10!}+\cdots$
$2^n = \frac{1}{0!}+\frac{n}{1!}+\frac{n(n-1)}{2!}+\frac{n(n-1)(n-2)}{3!}+\cdots$
$\frac{\pi}{4} = \frac{1}{1}-\frac{1}{3}+\frac{1}{5}-\frac{1}{7}+\frac{1}{9}-\frac{1}{11}+\frac{1}{13}-\frac{1}{15}+\frac{1}{17}-\frac{1}{19}+\frac{1}{21}-\frac{1}{23}+\frac{1}{25}\cdots$
$\infty = \frac{2}{1}\,\frac{4}{3}\,\frac{6}{5}\,\frac{8}{7}\,\frac{10}{9}\,\frac{12}{11}\,\frac{14}{13}\,\frac{16}{15}\,\frac{18}{17}\,\frac{20}{19}\,\frac{22}{21}\,\frac{24}{23}\,\frac{26}{25}\,\frac{28}{27}\,\frac{30}{29}\,\frac{32}{31}\,\frac{34}{33}\,\frac{36}{35}\cdots$
$0 = \frac{2}{3}\,\frac{4}{5}\,\frac{6}{7}\,\frac{8}{9}\,\frac{10}{11}\,\frac{12}{13}\,\frac{14}{15}\,\frac{16}{17}\,\frac{18}{19}\,\frac{20}{21}\,\frac{22}{23}\,\frac{24}{25}\,\frac{26}{27}\,\frac{28}{29}\,\frac{30}{31}\,\frac{32}{33}\,\frac{34}{35}\,\frac{36}{37}\cdots$
$\frac{\pi}{2} = \prod_{n=1}^{\infty}\left(1+\frac{1}{n}\right)^{(-1)^{n+1}} = \prod_{n=1}^{\infty}\left(\frac{2n}{2n-1}\cdot\frac{2n}{2n+1}\right) = \left(\frac{2}{1}\cdot\frac{2}{3}\right)\left(\frac{4}{3}\cdot\frac{4}{5}\right)\cdots$
$\frac{L}{2} = \prod_{n=1}^{\infty}\left(1+\frac{1}{2n}\right)^{(-1)^{n+1}} = \prod_{n=1}^{\infty}\left(\frac{4n-1}{4n-2}\cdot\frac{4n}{4n+1}\right) = \left(\frac{3}{2}\cdot\frac{4}{5}\right)\left(\frac{7}{6}\cdot\frac{8}{9}\right)\cdots$

$n! = \int_0^\infty(x-1)^n e^{-x}dx$
$!n = n!\sum_{k=0}^{n}\frac{(-1)^k}{k!}$
$!0 = 1$
$!1 = 0$
$\Gamma(s) = \int_0^\infty\frac{x^{s-1}}{e^x}dx$
$Q(1,x) = e^{-x}$
$P(1,x) = 1-e^{-x}$
$n! = \Gamma(n+1)$
$\Gamma(n) = (n-1)!$
$\frac{1}{\Gamma(-n)} = 0$
$\zeta(-2n) = 0$
$\zeta(s) = Li_s(1)$
$\zeta(0) = -\frac{1}{2}\log(e)$
$\zeta'(0) = -\frac{1}{2}\log(2\pi)$
$\zeta(s) = \frac{1}{\Gamma(s)}\int_0^\infty\frac{x^{s-1}}{e^x-1}dx$
$\zeta(s) = \prod_{p\text{ prime}}\frac{1}{1-p^{-s}}$
$\log(ab) = \log(a)+\log(b)$
$\log(n) = \int_1^n\frac{1}{x}dx$
$\frac{d}{dx}\log(x) = \frac{1}{x}$
$\log(x) = -\log\left(\frac{1}{x}\right)$
$Li_1(x) = -\log(1-x)$
$Li_0(x) = \frac{x}{1-x}$
$Li_{-1}(x) = \frac{x}{(1-x)^2}$
$Li_{-2}(x) = \frac{x(1+x)}{(1-x)^3}$

$\binom{n}{k} = \frac{n!}{k!(n-k)!}$
$\binom{2n}{n} = \frac{(2n)!}{(n!)^2} = \prod_{k=1}^{n}\frac{n+k}{k}$
$(a+b)^n = \sum_{k=0}^{n}\binom{n}{k}a^k b^{n-k}$

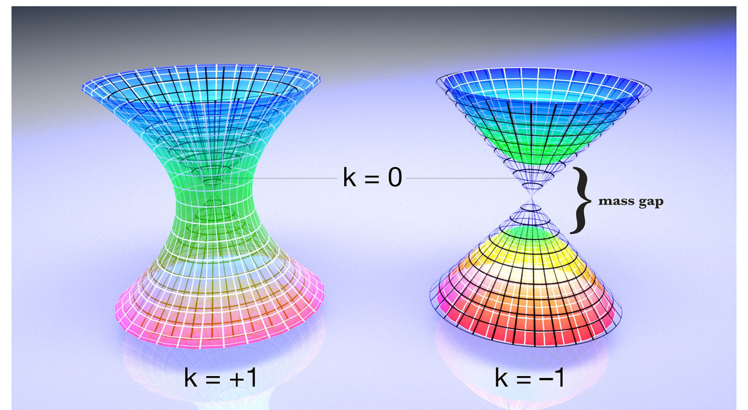
$|i| = 1 \qquad P+1 = P^3$
$|\varphi_i| = 1 \qquad P-1 = P^{-4}$
$i^4 = 1 \qquad \varphi = \frac{a}{b} = \frac{a+b}{a}$
$\varphi_i^6 = 1$
$i = (-1)^{1/2} \qquad \varphi_i = \frac{a}{b} = \frac{a-b}{a}$
$\varphi_i = (-1)^{1/3} \qquad \varphi+1 = \varphi^2$
$i = e^{\pi i/2} \qquad \varphi-1 = \frac{1}{\varphi}$
$\varphi_i = e^{\pi i/3} \qquad \varphi_i-1 = -\frac{1}{\varphi_i}$
$\varphi = e^{\pi i/5}+e^{-\pi i/5}$
$\sqrt{5} = \frac{1}{\varphi}+\varphi$
$F(n) = \frac{\varphi^n-(-\varphi)^{-n}}{\sqrt{5}}$
$\varphi = \lim_{n\to\infty}\frac{F_{n+1}}{F_n}$
$\varphi^n = \varphi^{n-1}+\varphi^{n-2}$
$\varphi_i^n = \varphi_i^{n-1}-\varphi_i^{n-2}$
$\varphi_i = \frac{\omega_1}{\omega_2}$
$Re(\omega_1) = \frac{1}{2}\omega_2$
$Im(\omega_1) = \frac{\sqrt{3}}{2}\omega_2$

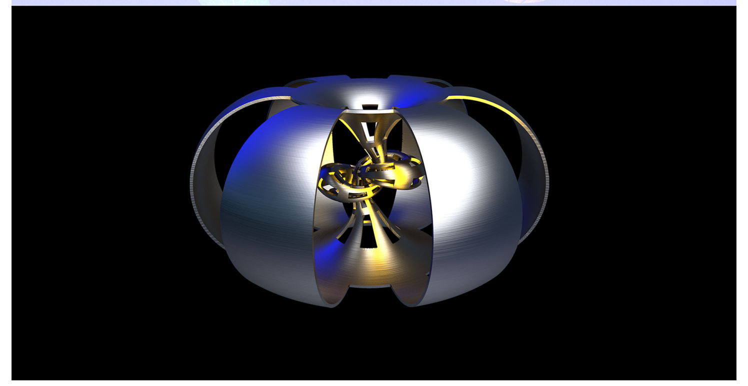
$\varphi = \sqrt{1+\sqrt{1+\sqrt{1+\sqrt{1+\cdots}}}}$
$\varphi_i = \sqrt{-1+\sqrt{-1+\sqrt{-1+\sqrt{-1+\cdots}}}}$
$P = \sqrt[3]{1+\sqrt[3]{1+\sqrt[3]{1+\sqrt[3]{1+\cdots}}}}$
$\varphi = \sqrt{1^2+\sqrt{1^2+\sqrt{1^2+\sqrt{1^2+\cdots}}}}$
$\varphi_i = \sqrt{i^2+\sqrt{i^2+\sqrt{i^2+\sqrt{i^2+\cdots}}}}$

$\cosh(x)$ = hyperbolic cosine
$\sinh(x)$ = hyperbolic sine
$\cos(x)$ = cosine
$\sin(x)$ = sine
$\tan(x)$ = tangent
$\cot(x)$ = cotangent
$\tan^{-1}(x)$ = inverse tangent
$\log(x)$ = natural logarithm
$Li_n(x)$ = polylogarithm of order n
$\Gamma(x)$ = gamma function
$Q(a,x)$ = Q-regularized gamma function
$P(a,x)$ = P-regularized gamma function
$\Psi(x)$ = digamma function
$\zeta(x)$ = Riemann zeta function
$\zeta'(x)$ = 1st derivative of Riemann zeta function
$B(x)$ = beta function
$B'(x)$ = 1st derivative of beta function
$Cl_s(x)$ = Clausen function
$K(i)$ = complete elliptic integral, 1st kind
$Im(x)$ = imaginary part of x
$Re(x)$ = real part of x
$!n$ = derangement
$n!$ = factorial
$\int$ = integral
$\sum$ = sum
$\prod$ = product
r_e = electron radius
r_+ = proton radius
r_n = neutron radius

Physics Monastery
Thad Roberts
September 23, 2025

$Cl_2(x) = \sum_{n=1}^{\infty}\frac{\sin(nx)}{n^2} = \frac{\sin(1x)}{1^2}+\frac{\sin(2x)}{2^2}+\frac{\sin(3x)}{3^2}+\frac{\sin(4x)}{4^2}+\frac{\sin(5x)}{5^2}+\cdots$
$Cl_2\left(2\pi n\pm\frac{\pi}{3}\right) = \pm G_{Gi} \qquad Cl_2(n\pi) = 0 \qquad \frac{d}{dx}Cl_2(x) = Cl_1(x)$
$Cl_2\left(2\pi n\pm\frac{2\pi}{3}\right) = \pm\frac{2G_{Gi}}{3} \qquad \max = G_{Gi} \qquad \min = -G_{Gi}$
$Cl_2(-x) = -Cl_2(x) \qquad \int_0^{\pi}Cl_2{}^2(x)\,dx = \zeta(4)$





legend

$t_p = 5.39125836832313\ldots\times 10^{-44}$ s	Planck time
$l_p = 1.61625918175645\ldots\times 10^{-35}$ m	Planck length
$q_p = 1.87554596713962\ldots\times 10^{-18}$ C	Planck charge
$T_p = 1.41678698590795\ldots\times 10^{32}$ K	Planck temperature
$m_p = 2.17642683817579\ldots\times 10^{-8}$ kg	Planck mass
5	number of dimensions
7	break in scale symmetry
$\text{ж}_1 = 0.0854245431533304\ldots$	1st hyperbolic partition constant
$\text{ж}_2 = 3.66756753485501\ldots$	2nd hyperbolic partition constant
$\text{ж}_3 = -1.87649603900417\ldots + 4.06615262615972\ldots\,i$	3rd hpc
$\text{ж}_4 = -1.87649603900417\ldots - 4.06615262615972\ldots\,i$	4th hpc
$\text{ж}_r = 4.47826244916751\ldots$	hyperbolic radius constant
$\text{ж}_\theta = 2.00316562310924\ldots$	hyperbolic radian constant
$G_{Gi} = 1.01494160640965\ldots$	Gieseking's constant
$V_{fe} = 2.02988321281930\ldots$	figure eight knot hyperbolic volume
$K = 0.9159655941772190\ldots$	Catalan's constant
$C_d = 0.291560904030818\ldots$	domino tiling constant
$D_d = 1.79162281206959\ldots$	dimer constant
$\omega_1 = 0.764977018528596\ldots + 1.32497062714087\ldots\,i$	omega_1 constant
$\omega_2 = 1.529954037057192\ldots$	omega_2 constant
$P = 1.32471795724474\ldots$	plastic constant
$F_{FF} = 1.22674201072035\ldots$	Fibonacci factorial constant
$i^{i^{i^{\cdots}}} = 0.438282936727032\ldots + 0.360592471871385\ldots\,i$	power tower of i
$i = 0+1i$	imaginary unit
$\pi = 3.14159265358979\ldots$	Archimedes' constant
$P_{up} = 2.29558714939263\ldots$	universal parabolic constant
$W_{We} = 0.474949379987920\ldots$	Weierstrass constant
$s = 5.24411510858423\ldots$	arc length of the unit lemniscate
$L = 2.622057554292119\ldots$	lemniscate constant
$L_1 = 1.31102877714605\ldots$	1st lemniscate constant
$L_2 = 0.599070117367796\ldots$	2nd lemniscate constant
$C_U = 0.847213084793979\ldots$	ubiquitous constant
$G_{Ga} = 0.834626841674073\ldots$	Gauss's constant
$C_{CF} = 0.697774657964819\ldots$	continued fraction constant
$C_{R1} = 0.998136044598509\ldots$	Ramanujan's 1st continued fraction constant
$e = 2.71828182845904\ldots$	Euler's number
$\gamma = 0.577215664901532\ldots$	Euler-Mascheroni constant
$S = 0.822825249678847\ldots$	Sierpiński constant
$\Omega = 0.567143290409783\ldots$	omega constant
$L_{LL} = 0.662743419349181\ldots$	Laplace limit
$D_{Do} = 0.739085133215160\ldots$	Dottie number
$C_{CFP} = 1.19967864025773\ldots$	Real fixed point of the hyperbolic cotangent
$\varphi_i = 0.5 + 0.866025403784438\ldots\,i$	the imaginary golden ratio
$\varphi = 1.61803398874989\ldots$	golden ratio
$C_Q = 0.605443657196732\ldots$	QRS constant
$C_{QA} = 2.03816937970215\ldots$	associated QRS constant
$\alpha_F = 2.50290787509589\ldots$	alpha Feigenbaum constant
$\delta_F = 4.66920160910299\ldots$	delta Feigenbaum constant

with:
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